



Vidyavardhini's

College of Engineering & Technology

Vasai Road (W)

Department of Artificial Intelligence & Data Science

Laboratory Manual: 2026-27

Student Copy

Semester	V	Class	T.E.
Course Code	2015111		
Course Name	Statistics for Machine Learning and Data Science Lab		



Vidyavardhini's College of Engineering & Technology

Vision

To be a premier institution of professional education, fostering academic excellence, innovation, and skill enhancement to serve industry and society.

Mission

- M1. To impart quality professional education through effective teaching-learning practices and industry-relevant curricula.
- M2: To foster research, innovation, and lifelong learning through interdisciplinary learning and modern infrastructure.
- M3. To strengthen industry collaboration for enhancing employability, entrepreneurship, and experiential learning.
- M4. To nurture ethical values, professionalism, leadership, and social responsibility for serving industry and society.



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

Department Vision:

To foster proficient artificial intelligence and data science professionals, making remarkable contributions to industry and society.

Department Mission:

- To encourage innovation and creativity with rational thinking for solving the challenges in emerging areas.
- To inculcate standard industrial practices and security norms while dealing with Data.
- To develop sustainable Artificial Intelligence systems for the benefit of various sectors.

Program Specific Outcomes (PSOs):

PSO1: Analyze the current trends in the field of Artificial Intelligence & Data Science and convey their finding by presenting / publishing at a national / international forums.

PSO2: Design and develop Artificial Intelligence & Data Science based solutions and applications for the problems in the different domains catering to industry and society.

Program Educational Outcomes (PEOs):

PEO1: Provide a strong foundation in mathematics, science, and engineering to enable graduates to excel in AI & Data Science careers and pursue higher studies.

PEO2: Equip graduates with the ability to apply modern tools and techniques to solve real-world problems in AI & Data Science.

PEO3: Develop ethical professionals with leadership qualities, social responsibility, and a commitment to lifelong learning.



Program Outcomes (POs):

Engineering Graduates will be able to:

- **PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop to the solution of complex engineering problems.
- **PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
- **PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.
- **PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.
- **PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
- **PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
- **PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
- **PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- **PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- **PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- **PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.



Lab Objectives

1	Apply descriptive statistics and sampling to analyze and visualize data.
2	Perform estimation, hypothesis testing, and resampling-based inference.
3	Implement regression and classification models and interpret diagnostic results
4	Apply unsupervised learning methods including PCA and clustering.
5	Evaluate ML models using appropriate metrics and cross-validation.
6	Interpret models with transparency, fairness, and ethical considerations

Lab Outcomes

CO	At the end of course students will be able to:	Action verbs	Bloom's Level
2015111.1	Perform exploratory data analysis using descriptive statistics and data visualization techniques to understand the characteristics of real-world datasets.	Perform	Apply (level 3)
2015111.2	Implement statistical inference methods, estimation techniques, hypothesis testing, and resampling methods for data-driven decision making.	Implement	Apply (level 3)
2015111.3	Develop regression models and evaluate their performance using appropriate statistical metrics and validation techniques.	Develop	Create (level 6)
2015111.4	Build and evaluate supervised machine learning models using appropriate classification algorithms and performance measures.	Build	Create (level 6)
2015111.5	Apply clustering and dimensionality reduction techniques to discover patterns and visualize high-dimensional datasets.	Apply	Apply (level 3)
2015111.6	Design an end-to-end statistical and machine learning solution by integrating data preprocessing, model development, evaluation, interpretation, and ethical considerations on a real-world dataset.	Design	Create (level 6)



Mapping of Experiments with Course Outcomes

List of Experiments	Course Outcomes					
	2015111 .1	201511 1.2	2015111 .3	2015111 .4	2015111 .5	201511 1.6
Perform Exploratory Statistical Analysis of a Real-World Dataset (Case Study)	3	-	-	-	-	-
Apply Descriptive Statistics and Data Visualization Techniques for Exploratory Data Analysis	3	-	-	-	-	-
Implement Correlation Analysis, Distance Measures, and Data Preprocessing Techniques	3	-	-	-	-	-
Perform Statistical Inference, Estimation, and Hypothesis Testing on Real-World Datasets	-	3	-	-	-	-
Implement Resampling Techniques and Confidence Interval Estimation for Statistical Decision Making	-	3	-	-	-	-
Develop and Evaluate Regression Models using Statistical Performance Measures	-	-	3	-	-	-



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Build and Evaluate Classification Models using Supervised Learning Algorithms	-	-	-	3	-	-
Apply Clustering and Dimensionality Reduction Techniques for Pattern Discovery	-	-	-	-	3	-
Evaluate and Interpret Machine Learning Models using Validation and Explainability Techniques	-	-	-	-	-	3
Design an End-to-End Statistical and Machine Learning Solution for a Real-World Problem (Course Project)	-	-	-	-	-	3



List of Experiments

Sr. No.	Name of Experiment	DOP	DOC	Marks	Sign
Basic Experiments					
1	Apply Descriptive Statistics and Data Visualization Techniques for Exploratory Data Analysis				
2	Implement Correlation Analysis, Distance Measures, and Data Preprocessing Techniques				
3	Perform Statistical Inference, Estimation, and Hypothesis Testing on Real-World Datasets				
4	Implement Resampling Techniques and Confidence Interval Estimation for Statistical Decision Making				
5	Develop and Evaluate Regression Models using Statistical Performance Measures				
6	Build and Evaluate Classification Models using Supervised Learning Algorithms				
7	Apply Clustering and Dimensionality Reduction Techniques for Pattern Discovery				
8	Evaluate and Interpret Machine Learning Models using Validation and Explainability Techniques				
Project / Case Study					
9	Perform Exploratory Statistical Analysis of a Real-World Dataset (Case Study)				
10	Design an End-to-End Statistical and Machine Learning Solution for a Real-World Problem (Course Project)				

D.O.P: Date of performance

D.O.C: Date of correction



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Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	
Roll No. :	
Experiment No.:	1
Title of the Experiment:	Perform Exploratory Statistical Analysis of a Real-World Dataset (Case Study)
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :



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Department of Artificial Intelligence & Data Science

Aim: To perform exploratory statistical analysis of the Pima Indians Diabetes Dataset by examining its structure, variable types, data quality, statistical characteristics, and important patterns.

Objectives

After completing this experiment, the student will be able to:

1. Apply exploratory data analysis techniques to understand the structure and characteristics of a real-world healthcare dataset.
2. Interpret the statistical characteristics, data quality issues, and important patterns identified during the analysis.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for female patients and is used to study factors associated with diabetes.

The dataset contains 768 observations, 8 input attributes, and one binary outcome variable.



Procedure

1. Load the Pima Indians Diabetes Dataset into a Pandas DataFrame.
2. Examine the dataset dimensions, column names, data types, and sample records.
3. Classify the variables as numerical, categorical, or binary.
4. Generate a statistical summary of the dataset.
5. Check for missing or invalid values.
6. Identify potential outliers using suitable statistical techniques and visualizations.
7. Generate appropriate visualizations to understand the data.
8. Record and interpret important observations obtained from the exploratory analysis.

Description of the Experiment

Exploratory Data Analysis (EDA) is the initial process of understanding a dataset before applying statistical inference or machine learning techniques.

The analysis follows: Raw Dataset, Data Inspection, Variable Identification, Data Quality, Analysis, Statistical Summary, Visualization and Interpretation

The Pima Diabetes Dataset is explored to understand its structure, data quality, variable distributions, and important characteristics before developing predictive models.

Detailed Description of Statistical techniques:

Descriptive Statistics

Statistical measures such as the following are used to summarize numerical variables:

- Mean
- Median
- Minimum
- Maximum
- Variance
- Standard Deviation

Missing Value Analysis

The dataset is examined for missing or invalid values.



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For example, zero values in certain medical attributes such as **Glucose, BloodPressure, SkinThickness, Insulin, and BMI** may require investigation because a zero value may represent missing or invalid data rather than a genuine medical measurement.

Outlier Analysis

Potentially unusual observations are identified using statistical summaries and visualizations such as boxplots.

An outlier should be investigated before deciding whether it should be removed or retained.

Data Visualization

Suitable visualizations are generated to understand:

- Variable distributions
- Frequency of diabetes outcomes
- Relationships between variables
- Potential outliers

Examples include:

- Histograms
- Boxplots
- Bar charts
- Scatter plots

Conclusion

Exploratory statistical analysis was performed on the Pima Indians Diabetes Dataset to understand its structure, variable characteristics, data quality, and important patterns. The analysis provides the foundation for further statistical analysis and machine learning model development.

Questions to be Answered by Students

1. Which variables in the dataset contain zero or potentially invalid values that require investigation?
2. Which variable contains potential outliers based on the analysis?
3. What are the important patterns observed between the input variables and the Outcome variable?
4. State **two important insights** obtained from the exploratory analysis.



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AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	
Roll No. :	
Experiment No.:	2
Title of the Experiment:	Apply Descriptive Statistics and Data Visualization Techniques for Exploratory Data Analysis
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

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Aim: To apply descriptive statistical measures and data visualization techniques to summarize, analyze, and interpret the distribution and relationships present in the Pima Indians Diabetes Dataset.

Objectives

After completing this experiment, the student will be able to:

- Apply descriptive statistical and visualization techniques to explore the characteristics of a real-world healthcare dataset.
- Interpret distributions, variability, relationships, and patterns obtained from statistical analysis and visualizations.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. Matplotlib
6. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for female patients and is used to study factors associated with diabetes.

The dataset contains 768 observations and 8 input attributes, along with a binary outcome variable indicating diabetes status.

Attribute	Description
Pregnancies	Number of pregnancies



Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin fold thickness
Insulin	2-Hour serum insulin
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes pedigree function
Age	Age in years
Outcome	1 = Diabetic, 0 = Non-diabetic

Procedure

1. Load the Pima Indians Diabetes Dataset into a Pandas DataFrame.
2. Inspect the dataset structure, dimensions, column names, and data types.
3. Calculate descriptive statistics including mean, median, mode, minimum, maximum, variance, and standard deviation for selected numerical variables.
4. Analyze the distribution of variables using histograms and boxplots.
5. Analyze the frequency of the Outcome variable using a bar chart.
6. Generate scatter plots to study relationships such as Glucose vs. BMI and Age vs. Glucose.
7. Generate a pair plot for selected numerical variables.
8. Interpret the statistical summaries and visualizations and record important observations.

Description of the Experiment

Descriptive statistics summarize the important characteristics of a dataset using numerical measures, while data visualization presents these characteristics graphically.

The experiment follows the workflow: Dataset, Descriptive Statistics, Distribution Analysis, Visualization, Relationship Analysis, Interpretation,
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Students will use the Pima Diabetes Dataset to analyze the central tendency, variability, distributions, and relationships among health-related variables.

Detailed Description of the Statistical techniques:

Mean, Median, and Mode

These measures describe the central or typical value of a dataset.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The median is the middle value, while the mode is the most frequently occurring value.

Range, Variance, and Standard Deviation

These measures describe data variability.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$s = \sqrt{s^2}$$

Histogram

Used to study the distribution, spread, and shape of numerical variables such as Glucose, BMI, and Age.

Boxplot

Used to visualize the median, quartiles, spread, and potential outliers.

$$IQR = Q_3 - Q_1$$



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Bar Chart

Used to display the frequency of categorical or binary variables such as Outcome.

Scatter Plot

Used to examine the relationship between two numerical variables, such as:

- Glucose and BMI
- Age and Glucose

Pair Plot

Used to simultaneously visualize pairwise relationships and distributions among multiple numerical variables.

Conclusion

Descriptive statistical measures and visualization techniques were applied to the Pima Indians Diabetes Dataset to understand the distribution, variability, and relationships among the variables.

Questions to be Answered by Students

1. Which numerical variable has the highest standard deviation?
2. Which variable shows the greatest difference between its mean and median?
3. What relationship can be observed between Glucose and BMI from the scatter plot?
4. State two important insights obtained from the analysis.
5. Which variable(s), if any, contain potential outliers? How were these identified?
6. State **two important insights** obtained from your exploratory statistical analysis of the dataset.



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AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	
Roll No. :	
Experiment No.:	3
Title of the Experiment:	Implement Correlation Analysis, Distance Measures, and Data Preprocessing Techniques
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

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Name of Faculty :

Signature :

Date :



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Aim: To implement correlation analysis, distance measures, and basic data preprocessing techniques on the Pima Indians Diabetes Dataset to identify relationships between variables and prepare data for further analysis.

Objectives

After completing this experiment, the student will be able to:

1. Apply correlation and distance measures to analyze relationships and similarities among data observations.
2. Implement basic data preprocessing techniques to prepare a dataset for statistical analysis and machine learning.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains medical diagnostic information for **768 female patients** and is used to study factors associated with diabetes.

The dataset contains numerical attributes and a binary outcome variable indicating whether the patient is diabetic.



Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Identify missing or invalid values, particularly zero values in medical attributes where zero may be unrealistic.
3. Handle selected missing values using an appropriate imputation technique.
4. Calculate the correlation matrix for numerical variables.
5. Visualize correlations using a heatmap.
6. Calculate Euclidean, Manhattan, and Cosine distances for selected observations.
7. Apply an appropriate normalization or standardization technique.
8. Compare the data before and after preprocessing.
9. Interpret the obtained correlation and distance measures.

Description of the Experiment

Correlation analysis helps determine the strength and direction of relationships between variables.

For example, the relationship between:

Glucose and BMI

can be analyzed using correlation.

Distance measures determine the similarity or dissimilarity between observations.

Data preprocessing is performed to improve the quality and consistency of data before further statistical or machine learning analysis.

The experiment follows: Raw Dataset, Identify Invalid/Missing Values, Data Preprocessing, Correlation Analysis, Distance Calculation and Interpretation

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Detailed Description of Statistical Methods

Correlation

Correlation measures the strength and direction of the linear relationship between two variables.

The Pearson correlation coefficient is:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

The value of r lies between:

$$-1 \leq r \leq 1$$

$r \approx 1$: Strong positive relationship

$r \approx -1$: Strong negative relationship

$r \approx 0$: Weak or no linear relationship

A correlation heatmap can be used to visualize relationships among multiple variables.

Euclidean Distance

Euclidean distance between two observations is:

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

It represents the straight-line distance between two points.

Manhattan Distance

Manhattan distance is:

$$d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n |x_i - y_i|$$

It measures distance by summing the absolute differences between corresponding features.



Cosine Distance

Cosine similarity is:

$$\cos(\theta) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

Cosine distance is:

$$d_{\cosine} = 1 - \cos(\theta)$$

It measures the difference in the direction of two vectors.

Data Preprocessing

Missing Value Handling

Invalid or missing values may be replaced using techniques such as:

- a. Mean imputation
- b. Median imputation

For medical variables, median imputation may be preferred when extreme values are present.

Normalization

Min-Max normalization transforms values into a specified range, usually [0, 1]:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Standardization

Standardization transforms a variable to have mean 0 and standard deviation 1:

$$z = \frac{x - \mu}{\sigma}$$

Scaling is important when distance-based methods are used because variables with larger numerical ranges can otherwise dominate the distance calculation.



Conclusion

Correlation analysis, distance measures, and data preprocessing techniques were implemented on the Pima Indians Diabetes Dataset. Correlation analysis helped identify relationships between medical variables, while distance measures quantified the similarity or dissimilarity between observations. Data preprocessing techniques helped improve the consistency and suitability of the data for further analysis and machine learning applications.

Questions to be Answered by Students

1. Which pair of variables has the strongest positive correlation in the dataset?
2. Which pair of variables has the strongest negative correlation, if any?
3. How do the Euclidean, Manhattan, and Cosine distances differ for the selected observations?
4. Which preprocessing technique was applied to the invalid or missing values, and why was it selected?
5. How does normalization or standardization affect the calculated distance between observations?



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AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	
Roll No. :	
Experiment No.:	4
Title of the Experiment:	Perform Statistical Inference, Estimation, and Hypothesis Testing on Real-World Datasets
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

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Name of Faculty :

Signature :

Date :



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

Aim: To apply statistical estimation and hypothesis testing techniques to the Pima Indians Diabetes Dataset and draw conclusions about population characteristics using sample data.

Objective

After completing this experiment, the student will be able to:

1. Apply estimation and hypothesis testing techniques to analyze population characteristics using sample data.
2. Interpret statistical test results and make data-driven conclusions based on the obtained evidence.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. SciPy
6. Matplotlib
7. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for 768 female patients

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select appropriate variables and divide the data into suitable groups where required.
3. Calculate point estimates such as sample mean and sample proportion.
4. Construct confidence intervals for selected population parameters.
5. Formulate null and alternative hypotheses for selected research questions.
6. Select and apply an appropriate statistical test, such as:



- a. One-sample t-test
 - b. Two-sample t-test
 - c. Chi-square test
7. Calculate the test statistic and p-value.
 8. Compare the p-value with the selected significance level.
 9. Make a statistical decision and interpret the result in the context of the dataset.

Description of the Experiment

Statistical inference allows conclusions about a population to be drawn using information from a sample.

The general process is: Sample Data, Estimate Population Parameter, Construct Confidence Interval, Formulate Hypotheses, Perform Statistical Test, and Interpret Results

For example, a sample of patients can be used to investigate whether the average glucose level differs from a specified reference value or whether diabetes outcome is associated with another categorical variable.

Detailed Description of Statistical Methods

Point Estimation

A sample statistic is used to estimate an unknown population parameter.

For example:

$$\bar{x}$$

is used to estimate the population mean μ .

A sample proportion is:

$$\hat{p} = \frac{x}{n}$$

where x is the number of observations with a particular characteristic.



Confidence Interval

A confidence interval provides a range of plausible values for a population parameter.

For a mean:

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

The confidence interval should be interpreted in the context of the population parameter being estimated.

Hypothesis Testing

A hypothesis test evaluates a claim about a population.

Null Hypothesis

$$H_0$$

represents the initial assumption.

Alternative Hypothesis

$$H_1$$

represents the claim being investigated.

A significance level such as:

$$\alpha = 0.05$$

may be used.

Decision Rule

- If $p\text{-value} < \alpha$, reject H_0 .
- If $p\text{-value} \geq \alpha$, fail to reject H_0 .



t-Test

A t-test can be used to compare means.

Examples:

- Test whether the mean glucose level differs from a specified value.
- Compare the mean BMI of two groups.

Chi-Square Test

The chi-square test can be used to test the association between categorical variables.

For example:

Diabetes Outcome and Gender

where applicable.

Conclusion

Statistical inference, estimation, and hypothesis testing techniques were applied to the Pima Indians Diabetes Dataset. Sample statistics were used to estimate population characteristics, confidence intervals were constructed, and hypothesis tests were performed to determine whether sufficient statistical evidence existed to support specific claims.

Questions to be Answered by Students

1. What population parameter was estimated in the experiment, and what was its estimated value?
2. What confidence interval was obtained, and how should it be interpreted?
3. What hypotheses were formulated for the selected statistical test?
4. Based on the p-value, what statistical decision was made?



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AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	
Roll No. :	
Experiment No.:	5
Title of the Experiment:	Implement Resampling Techniques and Confidence Interval Estimation for Statistical Decision Making
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

Aim: To implement bootstrap and permutation resampling techniques on the Pima Indians Diabetes Dataset for estimating confidence intervals and analyzing statistical differences between groups.

Objectives

After completing this experiment, the student will be able to:

1. Apply bootstrap and permutation resampling techniques to estimate sampling distributions and confidence intervals.
2. Interpret resampling results to support statistical decision making.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- SciPy
- Matplotlib
- Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for **768 female patients**. The Outcome variable indicates whether the patient is diabetic.



Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select a numerical variable such as Glucose or BMI.
3. Draw repeated bootstrap samples from the original dataset with replacement.
4. Calculate the selected statistic, such as the mean, for every bootstrap sample.
5. Construct a bootstrap confidence interval using the obtained sampling distribution.
6. Divide the data into two groups based on Outcome.
7. Calculate the observed difference between the group means.
8. Randomly permute the group labels repeatedly.
9. Calculate the difference in group means for every permutation.
10. Compare the observed difference with the permutation distribution and interpret the result.

Description of the Experiment

Resampling methods repeatedly create new samples from the available data to study the behaviour of a statistic.

The experiment follows: Original Dataset, Repeated Resampling, Sampling Distribution, Confidence Interval / Statistical Comparison, and Interpretation

Two resampling techniques are implemented:

- **Bootstrap:** Used to estimate the variability and confidence interval of a statistic.
- **Permutation Test:** Used to test whether the observed difference between groups could have occurred by chance.



Detailed Description of Statistical Methods

Bootstrap Resampling

Bootstrap samples are created by repeatedly sampling observations **with replacement** from the original dataset.

For each bootstrap sample, a statistic such as the mean is calculated.

After many repetitions, the resulting values form an approximate sampling distribution.

A confidence interval can then be obtained from this distribution.

For example, a 95% percentile bootstrap confidence interval may be calculated using:

$$[Q_{2.5}, Q_{97.5}]$$

where $Q_{2.5}$ and $Q_{97.5}$ are the 2.5th and 97.5th percentiles of the bootstrap distribution.

Permutation Test

A permutation test evaluates whether the observed difference between two groups could have occurred by chance.

For example:

$$\text{Difference in Mean Glucose} = \bar{x}_{diabetic} - \bar{x}_{non-diabetic}$$

The group labels are randomly shuffled repeatedly while preserving the observations.

The resulting differences form the **null distribution**.

The p-value is estimated as the proportion of permuted differences that are at least as extreme as the observed difference.



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Conclusion

Bootstrap and permutation resampling techniques were implemented on the Pima Indians Diabetes Dataset. Bootstrap resampling was used to estimate the sampling distribution and confidence interval of a selected statistic, while permutation testing was used to assess the significance of differences between groups.

Questions to be Answered by Students

1. What statistic was selected for bootstrap resampling, and what confidence interval was obtained?
2. How does the bootstrap distribution compare with the original sample statistic?
3. What was the observed difference between the selected groups?
4. Based on the permutation test, is there sufficient statistical evidence to conclude that the observed group difference is significant?
5. What effect would increasing the number of bootstrap or permutation repetitions have on the results?



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	
Roll No. :	
Experiment No.:	6
Title of the Experiment:	Develop and Evaluate Regression Models using Statistical Performance Measures
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

Aim: To develop regression models for predicting a continuous variable using the Pima Indians Diabetes Dataset and evaluate their performance using appropriate statistical metrics.

Objectives

After completing this experiment, the student will be able to:

1. Develop regression models to predict a continuous health-related variable from medical attributes.
2. Evaluate and interpret the performance of regression models using appropriate statistical measures.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains medical information for **768 female patients**.

For this experiment, **BMI may be selected as the continuous target variable**, while other relevant medical attributes are used as input features.



Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select a continuous target variable and relevant predictor variables.
3. Check and preprocess missing or invalid values.
4. Divide the dataset into training and testing sets.
5. Develop a suitable regression model.
6. Train the model using the training data.
7. Generate predictions for the test data.
8. Evaluate the model using appropriate performance metrics.
9. Analyze residuals and compare actual and predicted values.
10. Interpret the model results and identify possible limitations.

Description of the Experiment

Regression is a supervised learning technique used to predict a continuous numerical output.

The experiment follows: Dataset, Select Features and Target, Preprocess Data, Split Data, Train Regression Model, Generate Predictions, Evaluate Performance and Interpret Results

Students develop a regression model to estimate a continuous health-related variable based on other available attributes.

Detailed Description of Statistical Methods

Linear Regression

A simple linear regression model can be represented as:

$$y = \beta_0 + \beta_1 x + \epsilon$$



For multiple predictors:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

where:

- y = target variable
- x_i = predictor variables
- β_i = model coefficients
- ϵ = error term

Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

It represents the average absolute prediction error.

Mean Squared Error

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Larger errors are penalized more heavily.

Root Mean Squared Error

$$RMSE = \sqrt{MSE}$$

RMSE is expressed in the same units as the target variable.

Coefficient of Determination

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

R^2 indicates the proportion of variation in the target variable explained by the model.



Residual Analysis

A residual is:

$$e_i = y_i - \hat{y}_i$$

Residual analysis helps identify:

- Systematic prediction errors
- Nonlinear patterns
- Unequal variance
- Potential outliers

Conclusion

A regression model was developed to predict a continuous health-related variable using the Pima Indians Diabetes Dataset. The model was evaluated using appropriate regression metrics, and residual analysis was performed to understand the model's prediction errors and limitations.

Questions to be Answered by Students

1. Which variable was selected as the target, and why was it suitable for regression?
2. What are the values of MAE, MSE, RMSE, and R^2 obtained by the model?
3. Which performance metric provides the most useful interpretation for this problem?
Justify your answer.
4. What patterns or issues were observed from the residual analysis?
5. State **two limitations** of the developed regression model.



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Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	
Roll No. :	
Experiment No.:	7
Title of the Experiment:	Build and Evaluate Classification Models using Supervised Learning Algorithms
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

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Name of Faculty :

Signature :

Date :



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Department of Artificial Intelligence & Data Science

Aim: To build supervised classification models for predicting diabetes outcomes using the Pima Indians Diabetes Dataset and evaluate their performance using appropriate classification metrics.

Objectives

After completing this experiment, the student will be able to:

1. Build supervised classification models to predict the diabetes outcome of patients.
2. Evaluate and interpret classification model performance using appropriate evaluation metrics.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains medical information for **768 female patients**.

The target variable is:

- Outcome = 0: Non-diabetic
- Outcome = 1: Diabetic



Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Separate the input features and target variable.
3. Identify and handle invalid or missing values.
4. Divide the dataset into training and testing sets using a stratified split.
5. Apply suitable preprocessing, such as feature scaling where required.
6. Build one or more classification models, such as:
 - Logistic Regression
 - k-Nearest Neighbors
 - Decision Tree
7. Train the selected model(s) using the training data.
8. Generate predictions for the test data.
9. Construct the confusion matrix.
10. Calculate classification performance metrics.
11. Compare the obtained results and interpret the model performance.

Description of the Experiment

Classification is a supervised learning technique used to assign observations to predefined categories.

In this experiment, the objective is to classify a patient as: Patient Health Data, Classification Model gives Non-Diabetic and Diabetic.

The experiment follows: Dataset, Feature and Target Selection, Data Preprocessing, Train-Test Split, Classification Model, Prediction, and Performance Evaluation



Detailed Description of Statistical Methods

Logistic Regression

Logistic regression predicts the probability of a binary outcome.

The logistic function is:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

The predicted probability is converted into a class using a decision threshold.

Confusion Matrix

A confusion matrix contains:

Predicted Negative Predicted Positive

Actual Negative True Negative (TN) False Positive (FP)

Actual Positive False Negative (FN) True Positive (TP)

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Precision indicates how many predicted positive cases were actually positive.

Recall

$$Recall = \frac{TP}{TP + FN}$$

Recall indicates how many actual positive cases were correctly identified.



F1-Score

$$F1 = 2 \left(\frac{Precision \times Recall}{Precision + Recall} \right)$$

The F1-score balances precision and recall.

ROC-AUC

The ROC curve represents the relationship between:

- True Positive Rate
- False Positive Rate

The Area Under the Curve (AUC) provides an overall measure of the model's ability to distinguish between the two classes.

Conclusion

Supervised classification models were developed to predict diabetes outcomes using the Pima Indians Diabetes Dataset. The models were evaluated using the confusion matrix, accuracy, precision, recall, F1-score, and ROC-AUC. The experiment demonstrated that different evaluation metrics provide different perspectives on classification performance.

Questions to be Answered by Students

1. Which classification model was implemented, and what were its accuracy, precision, recall, and F1-score?
2. How many true positives and false negatives were obtained from the confusion matrix?
3. Which metric is more important for diabetes prediction: precision or recall? Justify your answer.
4. What was the ROC-AUC value, and what does it indicate about the model's performance?
5. State **two limitations** of the developed classification model.



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Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	
Roll No. :	
Experiment No.:	8
Title of the Experiment:	Apply Clustering and Dimensionality Reduction Techniques for Pattern Discovery
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

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Name of Faculty :

Signature :

Date :



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Aim: To apply clustering and dimensionality reduction techniques to the Pima Indians Diabetes Dataset to discover hidden patterns and visualize relationships among patients.

Objectives

After completing this experiment, the student will be able to:

1. Apply clustering and dimensionality reduction techniques to discover patterns in a high-dimensional dataset.
2. Interpret the resulting clusters and reduced-dimensional representations using suitable evaluation measures and visualizations.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains medical information for **768 female patients**. The Outcome variable indicates diabetes status.

The Outcome variable may be excluded during unsupervised learning and used later to compare the discovered clusters with the known diabetes categories.



Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select relevant numerical features for unsupervised analysis.
3. Handle invalid or missing values.
4. Standardize the selected features.
5. Apply K-Means clustering.
6. Select a suitable number of clusters using the Elbow Method.
7. Evaluate the clustering using the Silhouette Score.
8. Apply PCA to reduce the dimensionality of the dataset.
9. Visualize the observations in the reduced two-dimensional space.
10. Analyze the discovered clusters and interpret the results.

Description of the Experiment

Unsupervised learning identifies patterns in data without using a target variable during model development.

The experiment follows: High-Dimensional Dataset, Data Preprocessing, Feature Standardization, Clustering, Cluster Evaluation, Dimensionality Reduction, Visualization and Interpretation

Clustering groups similar observations, while dimensionality reduction represents high-dimensional data using fewer dimensions for visualization and analysis.

Detailed Description of Statistical Methods

K-Means Clustering

K-Means divides observations into k clusters by minimizing the distance between observations and their assigned cluster centroid.



The objective function is:

$$J = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

where:

- C_j = cluster j
- μ_j = centroid of cluster j

Elbow Method

The Elbow Method evaluates different values of k using within-cluster sum of squares.

The value of k is selected by identifying the point where adding more clusters provides diminishing improvement.

Silhouette Score

The Silhouette Score evaluates how well an observation fits within its assigned cluster.

$$s = \frac{b - a}{\max(a, b)}$$

where:

- a = average distance within the same cluster
- b = average distance to the nearest other cluster

A higher score generally indicates better-defined clusters.

Principal Component Analysis

PCA transforms correlated variables into a smaller number of uncorrelated principal components.



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The principal components are obtained from the covariance matrix using eigenvalues and eigenvectors.

The first principal components capture the maximum possible variance in the data.

PCA can be used to represent the dataset in two dimensions for visualization.

Conclusion

Clustering and dimensionality reduction techniques were applied to the Pima Indians Diabetes Dataset. K-Means clustering was used to group patients based on similarities in their medical attributes, while PCA was used to reduce the dimensionality of the dataset for visualization. The resulting clusters were evaluated and interpreted using appropriate measures and visualizations.

Questions to be Answered by Students

1. What value of k was selected using the Elbow Method, and why?
2. What Silhouette Score was obtained, and what does it indicate about the quality of the clusters?
3. Which variables contributed most to the first two principal components?
4. What patterns were observed in the PCA visualization of the clusters?
5. How do the discovered clusters compare with the known Outcome categories?



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AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	
Roll No. :	
Experiment No.:	9
Title of the Experiment:	Evaluate and Interpret Machine Learning Models using Validation and Explainability Technique
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

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Name of Faculty :

Signature :

Date :



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Aim: To apply model evaluation, Bayesian inference concepts, and explainable AI techniques to interpret and understand machine learning predictions on the Pima Indians Diabetes Dataset.

Objectives

After completing this experiment, the student will be able to:

1. Apply model evaluation and Bayesian concepts to analyze the reliability of machine learning predictions.
2. Apply explainable AI techniques to interpret model predictions and identify important features influencing the results.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- SHAP
- LIME

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for **768 female patients**.

The target variable is: Outcome = 0: Non-diabetic and Outcome = 1: Diabetic



Procedure

1. Load and preprocess the Pima Indians Diabetes Dataset.
2. Divide the dataset into training and testing sets.
3. Train a suitable classification model.
4. Evaluate the model using k-fold cross-validation.
5. Calculate suitable performance measures such as accuracy and F1-score.
6. Calculate the Brier Score and generate a calibration curve.
7. Explain selected model predictions using SHAP or LIME.
8. Identify the important features influencing the predictions.
9. Interpret and document the model evaluation and explainability results.

Description of the Experiment

A machine learning model should not be evaluated only on its performance on a single test split. Cross-validation provides a more reliable estimate of model performance.

The experiment follows: Dataset, Train Model, Cross-Validation, Performance Evaluation, Calibration Analysis, Explainable AI and Interpretation

The experiment combines three aspects:

- **Model Evaluation:** How well does the model perform?
- **Calibration:** How reliable are the predicted probabilities?
- **Explainability:** Why did the model make a particular prediction?

Detailed Description of Statistical Methods

k-Fold Cross-Validation

The dataset is divided into k subsets.



The model is trained on $k - 1$ subsets and tested on the remaining subset.

This process is repeated k times.

The final performance is calculated using the average performance across all folds.

Bias-Variance Trade-Off

A model should balance:

- **Bias:** Error due to an overly simple model.
- **Variance:** Error due to a model being too sensitive to the training data.

A good model attempts to achieve a suitable balance between bias and variance.

Calibration Curve

A calibration curve compares:

Predicted Probability

with:

Observed Frequency

A well-calibrated model produces probabilities that closely match the actual frequency of the outcome.

Brier Score

The Brier Score measures the accuracy of probabilistic predictions:

$$BS = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$$

where:

- p_i = predicted probability
- y_i = actual outcome



A lower Brier Score generally indicates better probabilistic predictions.

SHAP

SHAP values explain the contribution of individual features to a model prediction.

They help answer:

Which features contributed to increasing or decreasing the predicted probability?

LIME

LIME explains an individual prediction by creating small variations around the observation and approximating the model locally using an interpretable model.

Conclusion

The classification model was evaluated using cross-validation and calibration analysis to understand its performance and reliability. Explainable AI techniques such as SHAP or LIME were applied to identify the features influencing individual predictions. The experiment demonstrated that model performance, probability reliability, and interpretability are important aspects of evaluating machine learning systems.

Questions to be Answered by Students

1. What was the mean and standard deviation of the model performance across the cross-validation folds?
2. What was the obtained Brier Score, and what does it indicate about the reliability of the predicted probabilities?
3. Was the model well calibrated based on the calibration curve? Justify your answer.
4. Which features were identified as the most influential using SHAP or LIME?
5. Explain one individual prediction using the obtained explainability results.



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Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	
Roll No. :	
Experiment No.:	10
Title of the Experiment:	Design an End-to-End Statistical and Machine Learning Solution for a Real-World Problem (Course Project)
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

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Name of Faculty :

Signature :

Date :



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Department of Artificial Intelligence & Data Science

Aim: To design and implement an end-to-end statistical and machine learning solution for a real-world problem by integrating data preprocessing, statistical analysis, model development, evaluation, interpretation, and ethical considerations.

2. Objectives

After completing this experiment, the student will be able to:

1. Design and implement an end-to-end statistical and machine learning solution for a selected real-world problem.
2. Integrate data preprocessing, statistical analysis, model development, evaluation, interpretation, and ethical considerations into a complete solution.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Other relevant Python libraries, as required

Dataset Details

For this course project, students shall select a **real-world dataset** related to a suitable application domain such as:

- Healthcare
- Finance



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- Education
- Environment
- Social Media
- Business Analytics

The selected dataset should contain sufficient observations and relevant variables for performing statistical analysis and machine learning.

Students must clearly document:

- Dataset source
- Number of observations and variables
- Description of important attributes
- Target variable, if applicable
- Data quality issues

Procedure

1. Select a real-world problem and formulate the problem statement.
2. Identify and obtain a suitable dataset.
3. Perform data exploration and preprocessing.
4. Apply appropriate statistical analysis.
5. Select and develop a suitable machine learning model or models.
6. Evaluate the developed model using appropriate performance measures.
7. Interpret the model results using suitable visualization or explainability techniques.
8. Identify potential ethical, privacy, bias, or fairness concerns.



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9. Document the complete methodology, results, limitations, and conclusions.
10. Present the final solution and demonstrate the working implementation.

Description of the Experiment

The course project integrates the major concepts covered throughout the course.

Real-World Problem, Dataset Collection, Exploratory Data Analysis, Data Preprocessing, Statistical Analysis, Model Development, Model Evaluation, Interpretation, Ethical Analysis and Final Solution

Students are expected to make appropriate methodological choices based on the selected problem rather than following a fixed algorithm or dataset.

Detailed Description of Statistical and Machine Learning Methods

The methods used will depend on the selected problem.

Students may apply relevant techniques from the course, including:

Data Preprocessing

- Missing value handling
- Outlier treatment
- Encoding
- Normalization
- Standardization

Statistical Analysis

- Descriptive statistics
- Correlation analysis
- Confidence intervals



- Hypothesis testing
- Resampling techniques

Machine Learning

- Regression
- Classification
- Clustering
- Dimensionality reduction

Model Evaluation

Appropriate metrics should be selected based on the problem.

Examples include:

- MAE, MSE, RMSE, R^2
- Accuracy, Precision, Recall, F1-score
- ROC-AUC
- Silhouette Score

Interpretation and Ethics

Students should discuss:

- Important features influencing the results
- Model limitations
- Potential bias
- Data privacy concerns
- Fairness and transparency



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Conclusion

An end-to-end statistical and machine learning solution was designed and implemented for a selected real-world problem. The project integrated data preprocessing, exploratory analysis, statistical methods, model development, performance evaluation, interpretation, and ethical considerations.

Questions to be Answered by Students

1. What real-world problem was addressed, and why was it selected?
2. Why were the selected statistical and machine learning techniques appropriate for the problem?
3. What were the major findings and results obtained from the analysis?
4. What are the major limitations of the developed solution?
5. What ethical, privacy, bias, or fairness issues should be considered while applying the developed solution in the real world?

(Course Project format will be provided)